

Superconducting magnets for high field

EMFL User Meeting

University of Nottingham

11 June 2024

John Burgoyne

Head of Product Management

Key application drivers in high field superconducting magnets

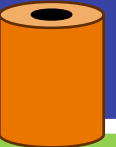
High field generation
for condensed matter
physics – “outsert”
magnets for HTS insert
coils



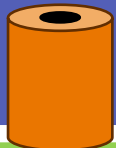
Conductor testing
(fusion and
accelerator-driven)



High field + ultra low
temperature physics –
including STM



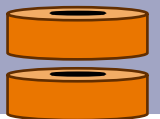
Dark matter – large
bore and/or high field
for axion searches



Materials processing

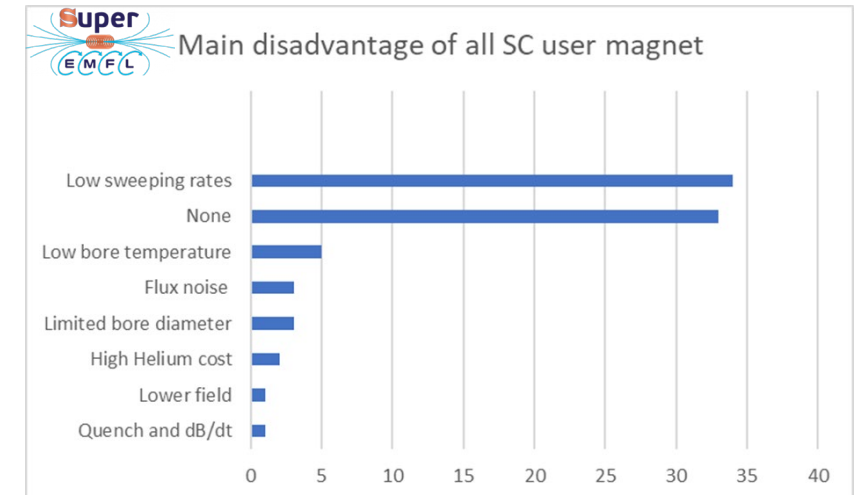
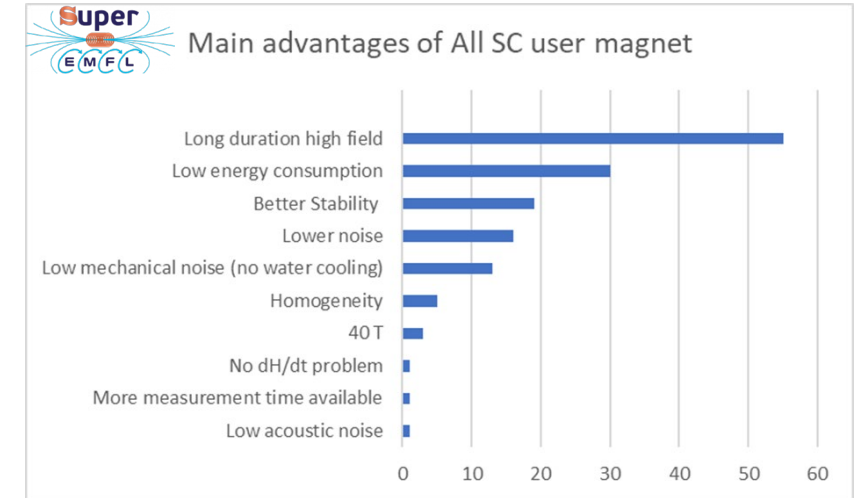


Beamlines –
particularly neutron
scattering



Why superconducting magnets?

- Experimental benefits
 - Lower noise, even assuming fully-driven mode >22 T
 - May unlock some effects at lower fields due to improved S:N
 - High homogeneity possible
 - Larger sample space sizes for e.g. DR integration
 - High uptime
 - In principle can be available 24/7
- Facility benefits
 - Compact
 - Relatively small physical footprint and easy service provision
 - Economic & sustainability balance
 - Low energy footprint balanced by liquid helium costs
 - Efficient helium recovery and reliquifier capability negates helium downside
- Downsides?



European high field installations

- University of Cambridge
 - 22 T* magnet system
 - Top-loading mK dilution refrigerator system
- UAM, Madrid
 - 22 T* magnet system
 - Dilution refrigerator capability
- HLD-HZDR, Dresden
 - 22 T* magnet system
 - 19 T, 150 mm bore magnet system
- University of Oxford
 - 21 T* magnet system



* At 2.2 K

High field all-superconducting user magnets

- National MagLab, Tallahassee, FL
 - 32 T all-superconducting magnet – world first
 - 15 T LTS + 17 T REBCO fully-insulated pancake coils
 - With top-loading mK dilution refrigerator capability
 - Typically running < 30 T as precaution for HTS coils
- Synergetic Extreme Condition User Facility (SECUF), Huairou (Beijing), China
 - Two high field systems
 - 28-30 T 15 T LTS + MI REBCO pancake coils
 - 26 T “NMR grade” 15 T LTS + BSCCO-2223 coils
 - With top-loading mK dilution refrigerator capability

NATIONAL HIGH
MAGNETIC
FIELD LABORATORY

User Facilities User Resources Research Magnet Development Education News & Events About Careers

Home / User Facilities / DC Field / Magnets & Instruments / Superconducting Magnets / SCM 32 T

32 Tesla Superconducting Magnet (SCM-32T)



The 32 T magnet is the first in a new class of high-field superconducting magnets employing HTS materials.

FLAGSHIP MAGNET

SCIENCE CHINA
Technological Sciences



•Letter•

<https://doi.org/10.1007/s11431-023-2611-9>

Achievement of a high-performance 30-tesla metal-as-insulated user superconducting magnet

ZHOU BenZhe^{1,2}, DAI YinMing¹, ZHANG ZiLi¹, LIU Hui¹, WANG YaoHui¹, WANG Lei¹,
WANG Hui¹, CHEN ShunZhong¹, WANG KangShuai^{1,2}, CHEN Yong^{1,2},
LIU JianHua^{1,2*} & WANG QiuLiang^{1,2*}

¹ Institute of Electrical Engineering, Chinese Academy of Sciences, Beijing 100190, China;
² University of Chinese Academy of Sciences, Beijing 100049, China

Received September 2, 2023; accepted January 29, 2024; published online May 22, 2024

Citation: Zhou B Z, Dai Y M, Zhang Z L, et al. Achievement of a high-performance 30-tesla metal-as-insulated user superconducting magnet. Sci China Tech Sci. 2024. Vol., <https://doi.org/10.1007/s11431-023-2611-9>

nature

View all journals

Search

Log in

Explore content

About the journal

Publish with us

Subscribe

Sign up for alerts

RSS feed

nature > news > article

NEWS | 29 April 2024

Superconductivity hunt gets boost from China's \$220 million physics 'playground'

From extreme cold to strong magnets and high pressures, the Synergetic Extreme Condition User Facility (SECUF) provides conditions for researching potential wonder materials.

By Gemma Conroy



The Synergetic Extreme Condition User Facility, just outside Beijing, can put samples through a battery of physical tests in one location. Credit: Institute of Physics, Chinese Academy of Sciences

- 2 x 22* T systems
- 2 x 20* T systems
- 20 T @ 4.2 K, 100 mm cold bore magnet system (SPM application)
- Multiple mK dilution refrigerator systems

* At 2.2 K

Superconducting high field projects for 30 T+

Europe

- EU Horizon project SuperEMFL, ending December 2024

- 30 T and 40 T goals
- Feasibility study only, 2.9 M€
- REBCO materials

- LNCMI “FASUM” project

- 40 T system build for EMFL, 4.3 M€



USA

- National Maglab

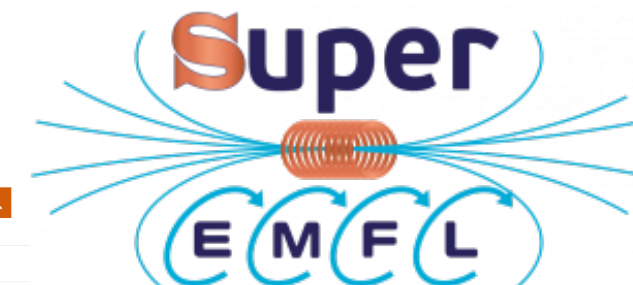
- 40 T feasibility study, \$M15.8
- REBCO materials
- Targeting development program from 2025/6 subject to funding

China

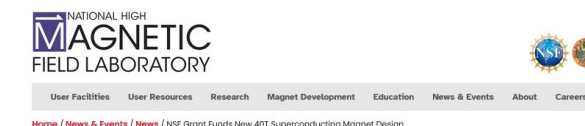
- CAS-IEE (Beijing) and IPP (Hefei) programs, \$M 100s?

Japan?

SuperEMFL project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement N°951714. Any dissemination of results reflects only the author's view and the European Commission is not responsible for any use that may be made of the information it contains.



Superconducting magnets for the European Magnetic Field Laboratory



NSF Grant Funds New 40T Superconducting Magnet Design

Published October 9/3, 2021



Designing on paper
Stephen Blesky

The world's next most powerful superconducting magnet will be designed at the National High Magnetic Field Laboratory.

More Stories

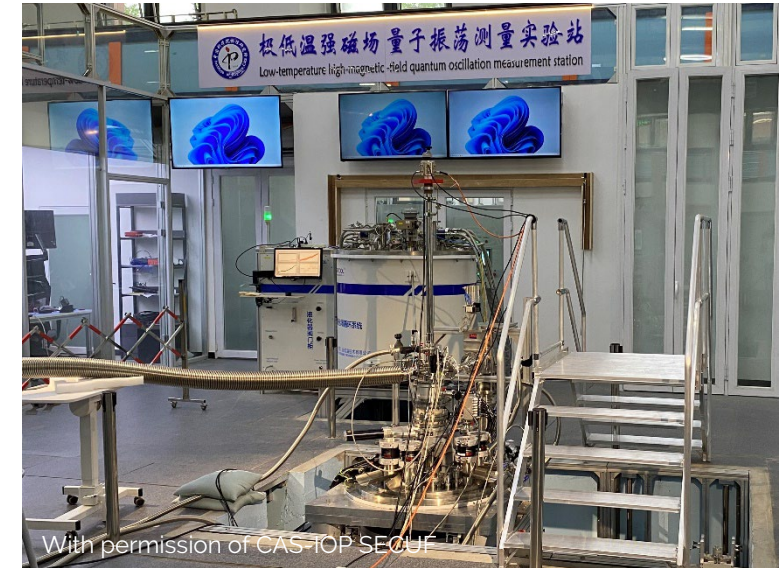
Wide-bore LTS non-persistent magnet technology



- 15 T, 250 mm LTS “outsert” for customer HTS insert coil
- Leveraging a large installed base of compact, high energy density magnets
- Based on design, modelling and manufacturing capabilities in:
 - Coil stress management
 - Quench management
 - Coil manufacturing and process control

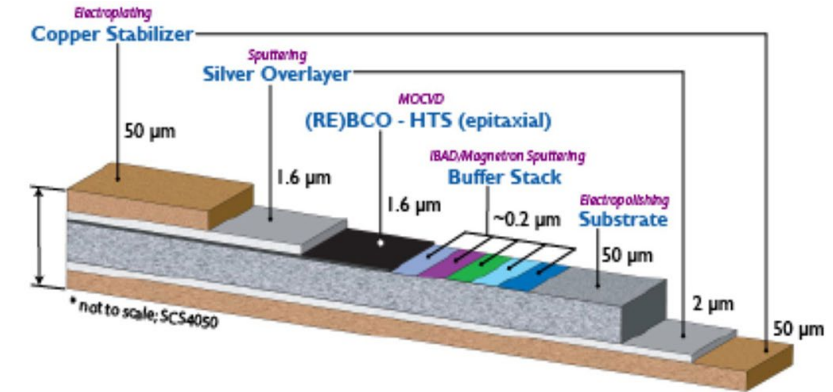
15 T / 250 mm
National Maglab
(NHMFL),
Tallahassee, FL
2014
→ 32 T

2 x 15 T / 250 mm
CAS-IOP “SECUF”,
Huairou, PRC
2021
→ 28-30 T and
26 T (NMR)

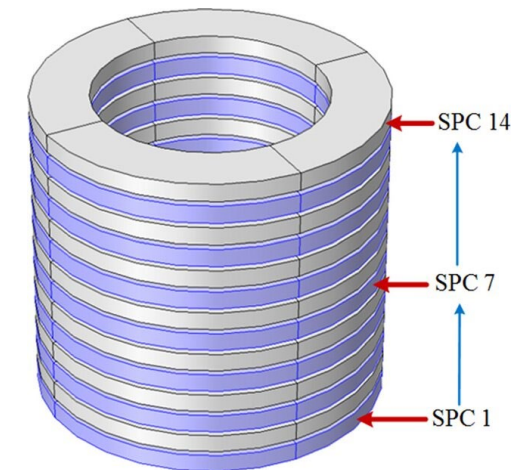


HTS conductor challenges for high field

- REBCO conductors
 - RE (= rare earth) - Ba - Cu - O
 - Coated tapes using pulsed laser deposition, sputtering, etc. – standard semiconductor/ industrial processes with in-process reaction
 - Multiple supply chain
 - Multiple insulation schemes – effectively different conductors
 - React-and-wind “pancake coils”
 - Complex quench scenarios
 - Development driven by **static field** applications
 - SuperEMFL 30 T & 40 T feasibility study



<https://nationalmaglab.org/magnet-development/applied-superconductivity-center/research/science-highlights/high-temperature-superconducting-tape/>

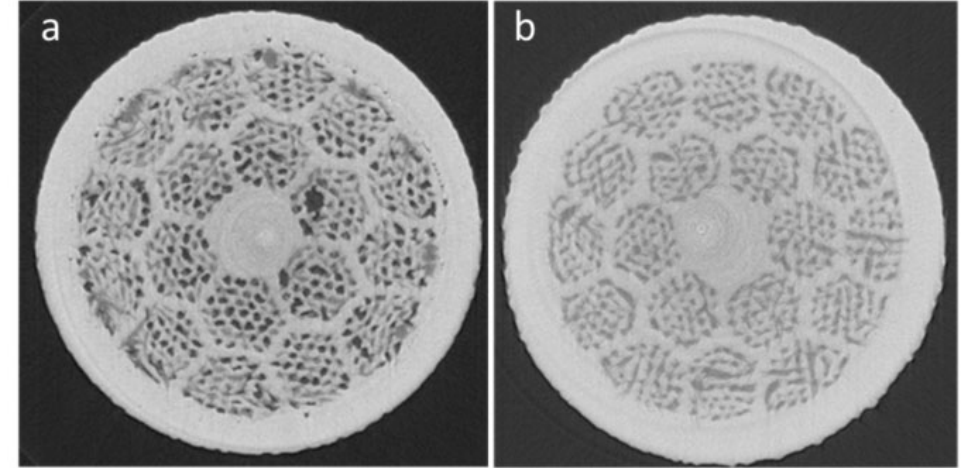


Wang et al, (2017), *J. Appl. Phys.*, **122**(5): 053902; DOI: 10.1063/1.4997738



HTS conductor challenges for high field

- BSCCO conductors
 - Bi – Sr – Ca – Cu – O
 - Powder-in-tube manufacture in Ag matrix, conventional wire-drawing & rolling processes
 - “2223” stoichiometry – flat tapes, used mostly in HTS current leads – very limited for high field magnets (but ref. SECUF and HTS-110)
 - “2212” stoichiometry – **round wire**, mainly developing via Applied Superconductivity Center (ASC), National Maglab, Tallahassee, FL
 - Limited supply chain
 - Wind-and-react layer-wound coils
 - Simplified quench scenarios
 - Behaves much more conventionally for ramped field applications



<https://nationalmaglab.org/magnet-development/applied-superconductivity-center/research-areas/bcco/>

1 March 2023

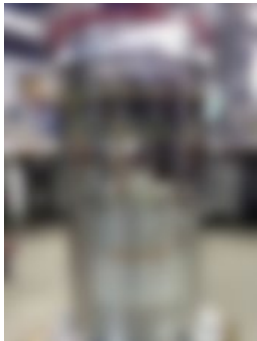
Oxford Instruments and MagLab's Applied Superconductivity Center Partner on High-Temperature-Superconducting Materials for High Field Research Magnets

<https://www.oxinst.com/news/news/oxford-instruments-and-maglab-applied-superconductivity-center-partnership/>

Wide-bore LTS persistent magnet technology

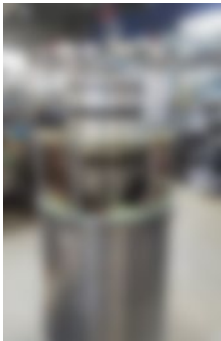


Central field @ 4.2 K (T)	Magnet bore size (mm)	Operation mode	Homogeneity over 10 mm DSV	Application	Country
18	150	Persistent	0.10%	STM	China
12	320	Persistent	< 0.02 %	Dark matter	S Korea
20	100	Persistent	< 0.10 %	STM	China
19.1 (1,300 T ² .m)	110	Persistent	< 0.10 %	Metallurgy	China



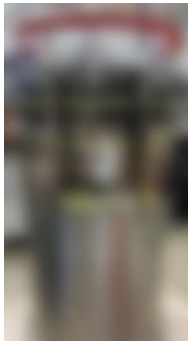
18 T / 150 mm

Persistent @ 4.2 K
2017



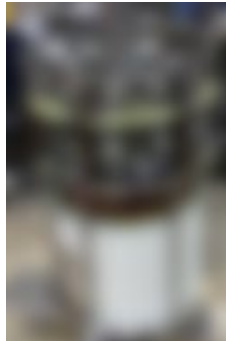
12 T / 320 mm

Persistent @ 4.2 K
2020



20 T / 100 mm

Persistent @ 4.2 K
2021



19 T / 110 mm / high gradient

Persistent @ 4.2 K
2022

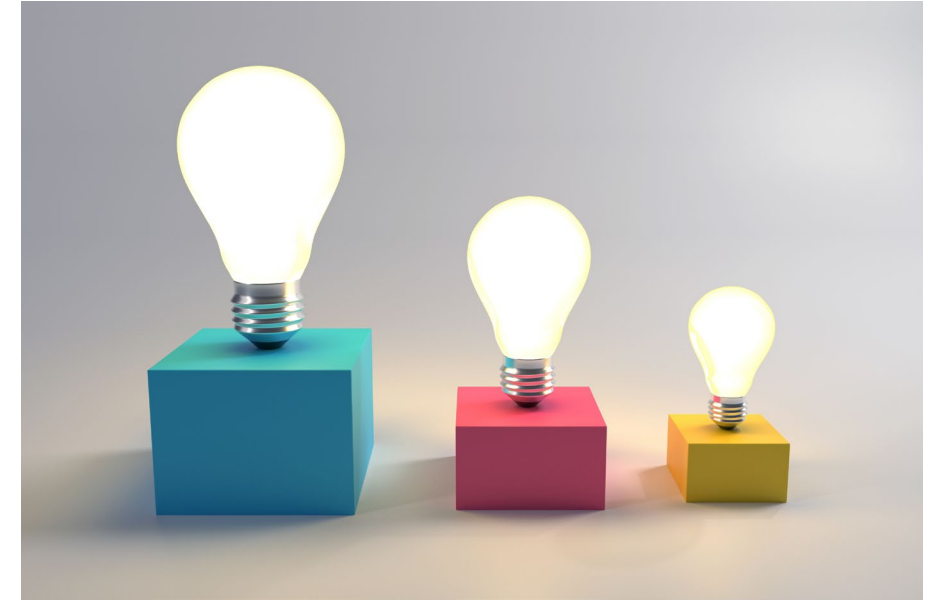
Commercial superconducting magnets

- Significant differences between high field magnets developed in e.g. national labs, and what can be a commercial offering
 - Predictability of performance
 - Cost & time for financial return
 - Guarantees
 - Commercial contract
 - Legal agreement of supply
 - Warranties (and reputation)
 - Sustain full operation & performance in the long term
 - There is (has to be!) a profit motive
 - Otherwise no re-investment in new products
 - Duty to shareholders, pension holders, employees



The outlook for high field all-superconducting magnets

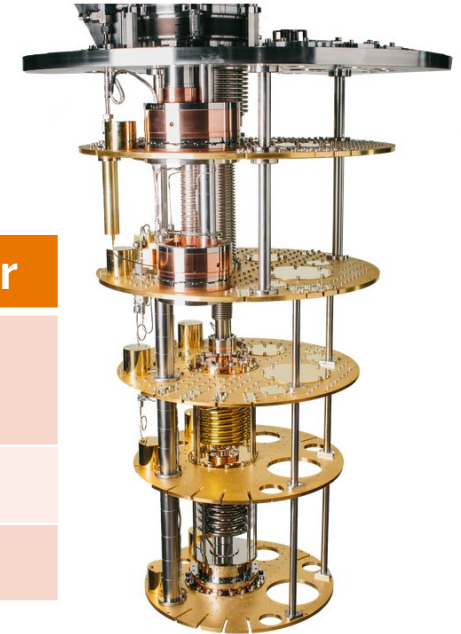
- Strong outlook for 30-40 T
- HTS materials are available and have sufficient performance
- Significant funding is needed
 - Europe is behind both US and China
 - FASUM *may* catch up?
- In-use issues have to be resolved, subject to HTS conductor selection
 - Quench
 - Ramp time
 - Repeatability of field setting



High B/T ratio – final thoughts

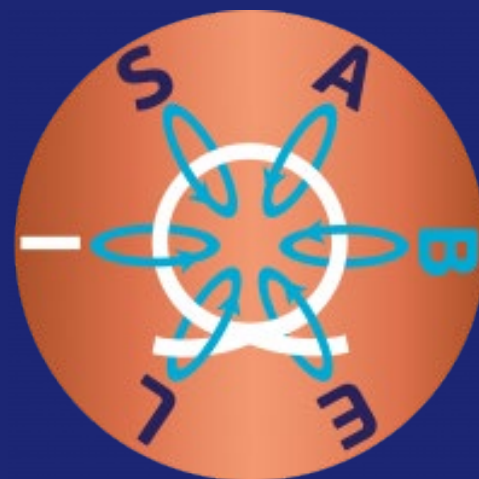
- If high B/T ratio is the critical experiments figure of merit...
- New, cryogen-free (“dry”) dilution refrigerators offer larger gains in T vs B

	B	T_{min}	T_{max}	CP, 100 mK	B/T factor
Typical resistive	40 T	25 mK	1 K?	~400 μ W	1,600
Wet	20 T (4.2 K)	15 mK	1 K	400 μ W	1,333
Dry	14 T	5 mK	30 K	850 μW	2,800



ISABEL

Thank you



Improving the **S**ustainability of the
European **M**magnetic **F**ield **L**aboratory



*ISABEL project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement N°870611.
Any dissemination of results reflects only the author's view and the European Commission is not responsible for any use that may be made of the information it contains.*

inclusive • innovative • trusted • purposeful